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### Hygiene system for a portable packaged food container

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#### Abstract

A system may automatically conduct hygiene cycles for a confined space such an interior space of a box suitable for storing and/or carrying packaged food. The system may include a hygiene device that can be attached to the box to conduct the hygiene cycles. The system may also automatically collect information related to the hygiene cycles and transmit the collected information to a network.

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## Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application is a continuation of U.S. patent application Ser. No. 17/656,566, filed May 25, 2022, which is a continuation of U.S. patent application Ser. No. 16/941,285, filed Jul. 28, 2020, now issued as U.S. Pat. No. 11,285,231, which is a continuation of U.S. patent application Ser. No. 16/349,413, filed May 13, 2019, now issued as U.S. Pat. No. 10,758,638, which is a U.S. National Stage Filing under 35 U.S.C. 371 from International Application No. PCT/CN2018/086468, filed on May 11, 2018, which applications are incorporated herein by reference in their entirety.

### TECHNICAL FIELD

(1) This document relates generally to device hygiene and more particularly to a method and apparatus for conducting automatic cleaning and/or sanitization cycles for a confined space such as interior of a box for containing packaged food.

### BACKGROUND

(2) Small and confined spaces, such as the interior of a food delivery box, are difficult to clean and/or sanitize. For example, boxes used to deliver ready-to-eat foods, such as ordered by a customer from a restaurant, need periodic cleaning and sanitization to ensure food hygiene. This may be done by manually spraying a chemical agent onto surfaces of each delivery box. Quality and thoroughness of such manual cleaning and/or sanitization depends on the person performing the procedure, and are inconsistent because, for example, there is no mechanism ensuring that the entire surfaces are covered and/or in a substantially uniform fashion. A two-step cleaning and sanitizing procedure makes each cycle even more complicated and less consistent. Additionally, a manual operation does not provide an objective record showing that a hygiene procedure is routinely followed.

### SUMMARY

(3) A system may automatically conduct hygiene cycles for a confined space such an interior space of a box suitable for storing and/or carrying packaged food. The system may include a hygiene device that can be attached to the box to conduct the hygiene cycles. The system may also automatically collect information related to the hygiene cycles and transmit the collected information to a network.

(4) An example of a system for cleaning and sanitizing a box configured to be carried by a user may include a hygiene device. The hygiene device may be configured to be attached to the box, and may include an agent container configured to contain a hygiene agent, a spray nozzle configured to be inserted into the box (or otherwise placed in the box to perform its function), a pump in fluid communication with the agent container and in fluid communication with the spray nozzle, and a control circuit. The pump may be configured to pump the hygiene agent through the spray nozzle to produce a mist of the hygiene agent in the box. The control circuit may be configured to control hygiene cycles and may include a communication circuit configured to transmit and receive information, a hygiene cycle management circuit configured to initiate each cycle of the hygiene cycles, and a monitoring circuit configured to record information related to performance of each cycle of the hygiene cycles and to transmit the recorded information out of the hygiene device through the communication circuit.

(5) An example of a method for cleaning and sanitizing a box configured to be carried by a user is also provided. The method may include verifying emptiness of the box, starting an automatic

hygiene cycle using a battery-powered portable hygiene device attached to the box, recording information related to performance of the hygiene cycle, and transmitting the recorded information to an analysis center for analysis and monitoring of hygiene status of the box.

(6) This summary is an overview of some of the teachings of the present application and not intended to be an exclusive or exhaustive treatment of the present subject matter. Further details about the present subject matter are found in the detailed description and appended claims. The scope of the present invention is defined by the appended claims and their legal equivalents.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a block diagram illustrating an embodiment of a system for cleaning and sanitizing a box.

(2) FIG. 2 is a block diagram illustrating an embodiment of a hygiene device of the system of FIG. 1.

(3) FIG. 3 is a diagram illustrating of an embodiment of portions of the hygiene device as seen from a rear view of the box.

(4) FIG. 4 is a diagram illustrating of an embodiment of a bottom of the box configured to accommodate a spray nozzle of the hygiene device.

(5) FIG. 5 is a diagram illustrating of an embodiment of portions of the hygiene device as seen on a cross-sectional view of the box.

(6) FIG. 6 is a diagram illustrating of an embodiment of a controller unit of the hygiene device.

(7) FIG. 7 is a block diagram illustrating an embodiment of a control circuit of the hygiene device.

(8) FIG. 8 is a flow chart illustrating an embodiment of a method for cleaning and sanitizing a box.

(9) FIG. 9 is a diagram illustrating an embodiment of a user interface device of the system of FIG. 1.

### DETAILED DESCRIPTION

(10) The following detailed description of the present subject matter refers to subject matter in the accompanying drawings which show, by way of illustration, specific aspects and embodiments in which the present subject matter may be practiced. These embodiments are described in sufficient detail to enable those skilled in the art to practice the present subject matter. References to “an”, “one”, or “various” embodiments in this disclosure are not necessarily to the same embodiment, and such references contemplate more than one embodiment. The following detailed description is demonstrative and not to be taken in a limiting sense. The scope of the present invention is defined by the appended claims, along with the full scope of legal equivalents to which such claims are entitled.

(11) This document discusses, among other things, a system for automatically conducting hygiene cycles for a confined space and automatically logging the hygiene cycles. The system can be constructed as a portable clean-in-place (“CIP”) system. The hygiene cycle log can include various information collected by the system and can be transmitted to an analysis center through a telecommunication network.

(12) In this document, “hygiene” can include cleaning, sanitization, or cleaning and sanitization (C&S). For example, a “hygiene agent” can include a cleaning agent, a sanitization agent, or a cleaning and sanitization (C&S) agent; a “hygiene cycle” can include a cleaning cycle, a sanitization cycle, or a cleaning and sanitization (C&S) cycle; and a “hygiene procedure” can include a cleaning procedure, a sanitization procedure, or a cleaning and sanitization (C&S) procedure.

(13) An example of the confined space includes the confined space in a closed box for carrying or storing packaged ready-to-eat foods. An example of such a box includes a food delivery box for



carrying cooked foods from a kitchen to a dining place in a place remote from the kitchen. A hygiene procedure for such a box may be established to ensure food safety (e.g., according to pertinent regulatory requirements). When the procedure is carried out manually, compliance is difficult to ensure, and no objective evidence of compliance is recorded.

(14) The present system provides an intelligent system that can spray a hygiene agent onto surfaces of a confined space with a substantially uniform distribution under automatic control for conducting hygiene cycles periodically or according to other specified schedule and/or criteria. In some embodiments, a 2-in-1 hygiene agent (also referred to as a C&S agent) combines cleaning and sanitation into an efficient single step. The present system also collects hygiene cycle information automatically and transmits the collected information to one or more designated receiving locations for analysis and monitoring.

(15) Existing hygiene systems that are intended for large spaces and need much power to operate, plenty of water to flush, and compress air to dry are not suitable for small spaces such as the food delivery box, which can be small enough, for example, to be carried by hand and/or by an electric moped or motorcycle. The present system can be battery-powered and portable, such as a system that can be easily attached to the food delivery box to perform controlled spray of the hygiene agent onto interior surfaces of the food delivery box when it is empty. In one embodiment, the present system uses a single spray nozzle configured to function with a diaphragm pump to spray the hygiene agent with nebulization to ensure that the hygiene agent covers all surfaces of a confined space, such as the cavity of a rectangular box having six planar surfaces.

(16) In various embodiments, the present system can include a control system that receives a command and initiates a hygiene cycle in response to the command. The command can be received by the system from a remote device via a communication link. In response to the hygiene cycle being initiated, an electronic circuit of the present system generates a signal to start the diaphragm pump to spray, resulting in a substantially uniform distribution of the hygiene agent on all the surfaces of the confined space. The electronic circuit can control automatic conduction of the hygiene cycle, record time of the hygiene cycle and/or other hygiene cycle information, and send the recorded information including the time to the analysis center.

(17) In various embodiments, the hygiene agent can include one or more chemical substances in liquid form. An example of the hygiene agent includes a cleaning agent, a sanitization agent, or a 2-in-1 C&S agent selected such that when being sprayed into a confined space using the present system, the quantity of the agent distributed on surfaces of the confined space can produce a result that satisfies pertinent regulatory requirements for cleaning and sanitizing efficacy. This quantity of agent distribution can be achieved by configuring the spray nozzle and diaphragm pump to provide for chemical nebulization and determining a duration of spraying for each hygiene cycle.

(18) While a hygiene system for the food delivery box is specifically discussed in this document as an example, the present subject matter is not limited to such an application, but can be used to apply liquid and/or gas into any confined space, for purposes including but not limited to cleaning and sanitizing the confined space. Some examples of devices having cavities that can be cleaned and sanitized by applying the present subject matter include like refrigerators, microwave ovens, food containers in any shape, containers of non-food items that need cleaning and sanitization, storage and/or delivery boxes (not limited to food storage and/or delivery), and wash machines. Additionally, the present subject matter is not limited to spraying liquid hygiene agents, but can also be used to spray various other liquid or gas such as pest control agents.

(19) FIG. 1 is a block diagram illustrating an embodiment of a system **100** for cleaning and sanitizing a box **101**. Box **101** can include any container having a cavity that needs to be cleaned and sanitized. In the illustrated embodiment, box **101** is a rectangular box with a flat, rectangular bottom **102** (only edges shown in FIG. 1), a lid (or cover) **103**, and four sides. The four sides include a front side **105** (only an edge shown in FIG. 1), a rear side **104**, and 2 lateral sides **106A** and **106B** (only an edge shown in FIG. 1) each coupled between front side **105** and rear side **104**.

Lid **103** is connected to rear side **104** with hinges **107A** and **107B**. In various embodiments, lid **103** is connected to rear side **104** with one or more hinges. Box **101** can be securely closed using a locking mechanism (not shown) that locks lid **103** in a closed position. When lid **103** is in the closed position (as shown in FIG. **1**), a confined space is formed by interior surfaces of box **101**, including interior surfaces of bottom **102**, front side **105**, rear side **104**, lateral sides **106A** and **106B**, and lid **103**. This confined space can be cleaned and sanitized by using the present subject matter.

(20) An example of box **101** is a food storage and/or delivery box such as a portable food delivery box. In various embodiments, box **101** as a food delivery box can be carried by hand and/or attached onto a vehicle such as a bicycle, a moped, or a motorcycle. For example, the food delivery box can be attached onto the vehicle with rear side **104** facing the forward-moving direction of the vehicle. In various embodiments, box **101** is used to contain packaged foods without direct contact between the foods and the interior surfaces of the box. When being used properly, no solid or liquid food waste or stain should remain on the interior surfaces of box **101** so that no rinse is needed for cleaning. Manual cleaning may be needed in case of an accidental food spill.

(21) System **100** can be an automatic hygiene system that includes a hygiene device **110**, a user interface device **120**, and an analysis center **124** that can communicate with user interface device **120** through a network **122**. Hygiene device **110** can be attached to box **101**. In some embodiments, hygiene device **110** is configured to be detachably attached to box **101**. In the illustrated embodiment, hygiene device **110** includes a container unit **111** to contain a hygiene agent, a controller unit **112** to control delivery of the hygiene agent to a spray nozzle (not shown in FIG. **1**), tubing **113** between container unit **111** and controller unit **112**, and tubing **114** between controller unit **112** and the spray nozzle. It should be understood that arrangement of system components (e.g., how the components of hygiene device **110** are physically arranged, housed, and attached to box **101**) as shown in FIG. **1** and other figures in this documents are each illustrated by way of example and not by way of restriction. Such arrangement can be determined based on design considerations such as a specific shape and size of box **101** and where box **101** is to be placed and/or how box **101** is to be transported.

(22) User interface device **120** is capable of transmitting and receiving information via one or more wireless communication links, such that it can present instructions to a user of box **101** and receive confirmation and/or commands from the user. In this document, a “user” includes a person who uses system **100** to clean and sanitize box **101**. When box **101** is a portable food delivery box, the user may be a delivery person (also referred to as “driver”, “rider”, etc.) who is assigned the task of cleaning and/or sanitizing the box. Such a delivery person can include any person carrying the box by hand and/or using any vehicle. User interface device **120** can communicate with hygiene device **110** via a wireless communication link **121** (e.g., using Bluetooth technology), and can communicate with network **122** via another wireless communication link **123** (e.g., the Internet through Wi-Fi, or a cellular network). In various embodiments, user interface device **120** can present the instructions to the user using a touchscreen and/or a speaker, and can receive the confirmation and/or commands from the user using the touchscreen and/or a microphone. For example, communicating with the user using the speaker and the microphone (e.g., a headphone set with a microphone) allows for hand-free operation at least to some extent. In various embodiments, user interface device **120** includes a mobile device such as a cellphone or a tablet computer.

(23) Analysis center **124** can represent one or more data analysis and/or monitoring facilities that communicate with many boxes of a business organization, such as all the food delivery boxes of a food delivery company, or a subset such as all the food delivery boxes within a region or a subsidiary of the food delivery company. This allows the organization to efficiently and effectively manage the cleaning and/or sanitization of the boxes, such as for ensuring food safety and/or meeting regulatory requirements.

(24) Network **122** can represent any network including, but not limited to, a telecommunication

network, a local network, the Internet, or the Internet of Things (IoT). In various embodiments, network **122** includes the IoT with user interface device **120** and analysis center **124** each being part of it. In some embodiments, hygiene device **110** can communicate with network **122** directly (without user interface device **120**) via another wireless communication link **127** (e.g., Wi-Fi or cellular network). This allows for communication between analysis center **124** or other operation control or management centers and hygiene device **110** directly when, for example, communication with user interface device **120** via links **121** and/or **123** is unavailable for any reason. In some embodiments, hygiene device **110** can be configured to be part of the IoT, and network **122** can represent the IoT, with hygiene device **110**, user interface device **120**, and analysis center **124** each being part of it.

(25) FIG. **2** is a block diagram illustrating an embodiment of a hygiene device **210**, which represents an example of components of hygiene device **110**. Hygiene device **210** can include an agent container **230**, tubing **213**, a pump **231**, tubing **214**, a spray nozzle **232**, a control circuit **233**, and a power source **234**.

(26) Agent container **230** contains the hygiene agent. Depending on the purpose of conducting hygiene cycles, the hygiene agent can be a liquid chemical agent that has cleaning effects only, sanitization effects only, or both cleaning and sanitization effects. A hygiene agent that has both cleaning and sanitization effects can be referred to as a 2-in-1 hygiene agent or 2-in-1 C&S agent. Using such a 2-in-1 hygiene agent allow each C&S hygiene cycle to be performed with a single spraying action. In various embodiments, agent container **230** includes a refillable or disposable bottle suitable for containing the hygiene agent. Pump **231** can receive the hygiene agent from agent container **230** through tubing **213** and deliver the hygiene agent to spray nozzle **232** through tubing **214**. An example of pump **231** includes a diaphragm pump. Spray nozzle **232** works with pump **213** to produce a mist of the hygiene agent in a confined space of a box such as box **101** to result in substantially uniform distribution of the hygiene agent on the interior surfaces of the box. In one embodiment as a specific example discussed in this document, spray nozzle **232** includes a single nozzle incorporated into the bottom of the box, such as by inserting into the box through a hole in a geometric center of the bottom of the box. In various embodiments, spray nozzle **232** can include one or more spray nozzles incorporated into the bottom, the lid, and/or each side of the four sides of the box to produce a mist of the hygiene agent in the box to result in substantially uniform distribution of the hygiene agent on the interior surfaces of the box. Control circuit **233** can include an electronic circuit configured to control conduction of the hygiene cycles. Power source **234** provides power for the operation of hygiene device **210**. Power source **234** can include one or more rechargeable and/or non-rechargeable batteries.

(27) In one embodiment, the components of hygiene device **210** are arranged and attached to the box as illustrated in FIG. **1**, with container unit **111** including agent container **230**, tubing **113** corresponding to tubing **213**, controller unit **112** including pump **231**, control circuit **233**, and power source **234**, tubing **114** corresponding to tubing **214**, and spray nozzle **232** installed at bottom **102**. In various embodiments, the components of hygiene device **210** can be arranged and coupled to the box in any way that ensures portability of the box.

(28) FIG. **3** is a diagram illustrating of an embodiment of portions of hygiene device **110** as seen from a rear view of box **101**. Components of box **101** that are shown in FIG. **3** include bottom **102** (only an edge shown), lid **103** (only an edge shown), lateral sides **106A** (only an edge shown) and **106B** (only an edge shown), and rear side **104**. Components of hygiene device **110** that are shown in FIG. **3** include container unit **111**, tubing **113**, controller unit **112**, and tubing **114**.

(29) FIG. **4** is a diagram illustrating of an embodiment of bottom **102** of box **101** configured to accommodate a spray nozzle. Components of box **101** that are shown in FIG. **4** include bottom **102**, lid **103** (only two edges shown), rear side **104**, front side **105** (only two edges shown), and lateral sides **106A** (only two edges shown) and **106B**. Bottom **102** includes a hole **440** sized and shaped to allow placement of the spray nozzle such as spray nozzle **232**. Bottom **102** also includes a groove

**441** to accommodate a portion of tubing **114** that is connected to the spray nozzle.

(30) FIG. 5 is a diagram illustrating of an embodiment of portions of hygiene device **110** as seen on a cross-sectional view of box **101**. Components of box **101** that are shown in FIG. 5 include bottom **102** with hole **440** and groove **441**, lid **103**, and lateral sides **106A** and **106B**. Components of hygiene device **110** that are shown in FIG. 3 include container unit **111**, tubing **113**, tubing **114**, and a spray nozzle **532** representing an example of spray nozzle **232** and shown as producing a mist distributing the hygiene agent onto the interior surfaces of box **101**.

(31) FIG. 6 is a diagram illustrating of an embodiment of a controller unit **612**, which represent an example of controller unit **112**. Controller unit **612** as illustrated in FIG. 6 shows an example of placement of its major components including a pump **631** (representing an example of pump **231**), a control circuit **633** (representing an example of control circuit **233**), a battery pack **634** (representing an example of power source **234**), a tubing connector **643** configured to connect to tubing **113**, and a tubing connector **644** configured to connect to tubing **114**.

(32) FIG. 7 is a block diagram illustrating an embodiment of a control circuit **733**, which represent an example of control circuit **233**. Control circuit **733** can include a communication circuit **750**, a power switch **751**, a hygiene cycle management circuit **752**, a monitoring circuit **753**, and a power management circuit **754**.

(33) Communication circuit **750** provides for communication between user interface device **120** and hygiene device **110** via wireless communication link **121**, such as using Bluetooth technology. In various embodiments, communication circuit **750** transmits data collected by hygiene device **110** to user interface device **120**, receives commands controlling operation of hygiene device **110** from user interface device **120**, and transmits signals indicative of operational status of hygiene device **110** to user interface device **120**. User interface device **120** can function as a relay device for communications between hygiene device **110** and analysis center **124** (or any other operation control or management center) through network **122**. In some embodiments, communication circuit **750** also provides for communication between network **122** and hygiene device **110** via wireless communication link **123**, such as using Wi-Fi or a cellular network. Communication circuit **750** can transmit data collected by hygiene device **110** to analysis center **124**, receive commands controlling operation of hygiene device **110** from analysis center **124**, and transmit signals indicative of operational status of hygiene device **110** to analysis center **124**, through network **122** without using user interface device **120** as a relay device.

(34) Power switch **751** can include any switch suitable for turning electric power on and off. This allows the user to turn hygiene device **110** on and off.

(35) Hygiene cycle management circuit **752** can initiate and control a hygiene cycle according to programmed schedule, criteria, and/or user command. In various embodiments, hygiene cycle management circuit **752** can be configured to initiate a hygiene cycle periodically (e.g., every 4 hours) or according to a specified (e.g. programmable) schedule (e.g., at 10:30 am, 2:00 pm, 5:00 pm, and 8:00 pm). Hygiene cycle management circuit **752** can also be configured to initiate a hygiene cycle in response to a user command (e.g., overriding the specified schedule). The user command can include a command transmitted from user interface device **120** and received by communication circuit **750**. In various embodiments, hygiene cycle management circuit **752** can confirm that box **101** is empty and that lid **103** is closed, and can start spraying the hygiene agent upon the confirmation. For example, hygiene cycle management circuit **752** can transmit a confirmation request signal through communication circuit **750** to user interface device **120**, and receive a confirmation response signal confirming that box **101** is empty and that lid **103** is closed from user interface device **120**. User interface device **120** can present the confirmation request signal as an audial and/or visual message to the user, and receive the response from the user as the confirmation response signal. The confirmation response signal can act as a command for starting the spraying of the hygiene agent, i.e., hygiene cycle management circuit **752** can start spraying the hygiene agent upon receiving the confirmation response signal. In various embodiments, hygiene

cycle management circuit **752** can also control the performance of the hygiene cycle following its initiation, including controlling the pumping of the hygiene agent during the hygiene cycle. For example, hygiene cycle management circuit **752** can control the duration of the pumping of the hygiene agent by timing the pumping and ending the pumping when a specified time interval expires.

(36) Monitoring circuit **753** can record information indicative of performance of each hygiene cycle and transmit the recorded information to network **122** (in other words, “upload data to the cloud”). In various embodiments, the recorded information can include information identifying box **101** (e.g., a serial number), the user (e.g., the delivery person using the box), time of the hygiene cycle (e.g., starting, ending, and/or actual spraying times), and/or location where the hygiene cycle is conducted. In various embodiments, the recorded information is transmitted to network **122** such that analysis center **124** can analyze the recorded information and send instructions and/or warnings to user interface device **120** based on an outcome of the analysis (e.g., to ensure compliance of hygiene procedures), receive recorded information from all the delivery boxes equipped with the hygiene devices (e.g., company-wide, region-wide, or nationwide) on a regular basis, report performance (e.g., statistical data) of the hygiene cycles for the delivery boxes to management (e.g., a designated person or team) periodically (e.g., monthly), and/or send instructions to each user and/or each team leader managing a plurality of users from the management. In various embodiments, monitoring circuit **753** can also monitor the status of the hygiene cycle and notify the user by transmitting information indicative the status of the hygiene cycle to user interface device **120**. When the transmitted information indicates that the hygiene cycle is completed, user interface device **120** can present an instruction for the user to open lid **103** to allow natural drying or to use paper towel to wipe the interior surfaces of box **101** upon completion of the hygiene cycle (e.g., for compliance with regulatory requirement) and/or to perform certain other tasks in compliance with pertinent regulatory requirements.

(37) Power management circuit **754** can manage the operation of power source **234** and control distribution of electric power to various components of hygiene device **110** that require the electric power to function (e.g., pump **231** and control circuit **233** or **733**). When power source **234** includes one or more batteries, power management circuit **754** can monitor the remaining level of the one or more batteries and produce a battery level alert signal in response to the level dropping under a battery level threshold. Power management circuit **754** can transmit this battery level alert signal through communication circuit **750** to user interface circuit **120** to remind the user to replace or recharge the one or more batteries of power source **234**. In one embodiment, the user interface device **120** reminds the user to replace or recharge the battery periodically (e.g., weekly).

(38) In some embodiments, control circuit **733** is configured to determine the location of hygiene device **110** using a positioning system such as the global positioning system (GPS) or location based services (LBS). This allows for tracking of hygiene device **110** and use of its location tracks to determine whether a hygiene cycle should be conducted. Hygiene cycle management circuit **752** can use the location of hygiene device **110** as one of the criteria to initiate the hygiene cycle. For example, when box **101** is due for a scheduled hygiene cycle, hygiene cycle management circuit **752** can track the location of the attached hygiene device **110** and initiate the hygiene cycle after box **101** with hygiene device **110** returns after a food delivery to a facility while the delivered foods are prepared. The information about the hygiene cycle recorded and transmitted by monitoring circuit **753** can also include the location where the hygiene cycle is conducted. Control circuit **733** and/or analysis center **124** can analyze the recorded information to determine whether the hygiene procedure is complied, for example, by determining the number of food deliveries (as monitored by tracking locations of hygiene device **110**) between two hygiene cycles. Control circuit **733** can also send information indicating location of hygiene device **110** to network **122** to allow for tracking of hygiene device **110** by analysis center **124** and/or other operation control or management centers for various other purposes. For example, the location of hygiene device **110** can be tracked for

recording and analyzing routes of food delivery to improve timing or efficiency, or to monitor whether hygiene device **110** is deployed in compliance of the hygiene procedure.

(39) FIG. **8** is a flow chart illustrating an embodiment of a method **860** for cleaning and sanitizing a box, such as box **101**. In one embodiment, system **100** is configured for performing method **860**. In one embodiment, a user interface device such as user interface device **120** is configured to control the performance of method **860**. For example, the user interface device can include a cellphone, and an application is installed on the cellphone for controlling the performance of method **860**. The application can present reminder and instructions to the user before, after, and during each hygiene cycle.

(40) At **861**, emptiness of the box is verified. If the box is not empty (e.g., food is contained in the box), the contents must be removed before a hygiene cycle can be started. If the box is not empty because it is dirty (e.g., resulting from food spill), the box can be manually cleaned (e.g., using a paper towel to wipe out solid or liquid food spills). Emptiness of the box can then be verified.

(41) At **862**, an automatic hygiene cycle is started in response to a command indicating the verification that the box is empty. The hygiene cycle can be automatically conducted using a battery-powered portable hygiene device attached to the box, such as hygiene device **110**.

(42) At **863**, information related to performance of the hygiene cycle is recorded by the battery-powered portable hygiene device. Such information indicates whether a hygiene procedure is complied, for example for ensuring food safety and/or meeting regulatory requirements.

(43) At **864**, the recorded information is transmitted to an analysis center, such as by uploading data representing the recorded information to a network. This concludes the hygiene cycle. In various embodiments, the transmission of the recorded information can be triggered by end of each hygiene cycle, on a periodic basis, by a data transmission command transmitted from the user interface device, and/or by a detection of failure in complying with the hygiene procedure. The recorded information can be transmitted to the analysis center from the battery-powered portable hygiene device directly and/or via the user interface device.

(44) In various embodiments, upon completion of the hygiene cycle, interior surfaces of the box are to be dried by the user. This can be done by opening the lid of the box to allow natural air dry. Alternatively, this can be done by opening the lid of the box and wiping all the interior surfaces of the box (e.g., using a paper towel), such as when time or weather does not allow for air dry. When the box is used for food delivery, no liquid (i.e., the hygiene agent) should remain in the box when loading packaged food for the next food delivery.

(45) In various embodiments, exterior of the box is also to be cleaned periodically. Such cleaning can be performed manually.

(46) In various embodiments in which the container of the hygiene agent is refillable, the container is to be cleaned before each refill or after a specified number of refills. Other parts of the hygiene device (e.g., the nozzle and/or the tubing) are also to be cleaned periodically or as needed.

(47) FIG. **9** is a diagram illustrating an embodiment of a user interface device **920**, which can represent an embodiment of user interface device **120**. In the illustrated embodiment, user interface device **920** is a cellphone on which a hygiene application **970** is installed. The cellphone is to be carried by each delivery person (or another person) responsible for compliance with the hygiene procedure.

(48) In various embodiments, functions performed by application **970** through cellphone **920** can include those selected from the following examples: presenting food safety tips for home delivery of cooked food; displaying a training video on a hygiene procedure for home delivery of cooked food; reminding the user that a hygiene cycle is starting (with an audial and/or visual message); instructing the user to check whether the box is empty; receiving confirmation from the user that the box is empty; triggering spray of the hygiene agent during each hygiene cycle; recording information related to each hygiene cycle; sending the recorded information to the network (uploading data representative of the recorded information to the analysis center through the

network as triggered by end of each hygiene cycle, on a periodic basis, by a transmission command from the user, and/or by a detection of failure in complying with a hygiene procedure; detecting each failure in complying with the hygiene procedure; receiving feedback information from the analysis center (the feedback information produced automatically by the data analysis center and/or by one or more persons such as the management); and presenting the feedback information (e.g., confirmation of compliance, reminder and/or instructions for actions by the user, and/or warnings related to each failure in complying with the hygiene procedure as detected by the application or the analysis center).

The above is not an exhaustive list, and other related functions may be included as determined by those skilled in the art.

(49) Some non-limiting examples (Examples 1-50) of the present subject matter are provided as follows:

(50) In Example 1, a system for cleaning and sanitizing a box configured to be carried by a user may include a hygiene device. The hygiene device may be configured to be attached to the box, and may include an agent container configured to contain a hygiene agent, a spray nozzle configured to be inserted into the box (or otherwise placed in the box to perform its function), a pump in fluid communication with the agent container and in fluid communication with the spray nozzle, and a control circuit. The pump may be configured to pump the hygiene agent through the spray nozzle to produce a mist of the hygiene agent in the box. The control circuit may be configured to control hygiene cycles and may include a communication circuit configured to transmit and receive information, a hygiene cycle management circuit configured to initiate each cycle of the hygiene cycles, and a monitoring circuit configured to record information related to performance of each cycle of the hygiene cycles and to transmit the recorded information out of the hygiene device through the communication circuit.

(51) In Example 2, the subject matter of Example 1 may optionally be configured such that the hygiene agent includes a cleaning agent.

(52) In Example 3, the subject matter of Example 1 may optionally be configured such that the hygiene agent includes a sanitization agent.

(53) In Example 4, the subject matter of Example 1 may optionally be configured such that the hygiene agent includes a cleaning and sanitization agent.

(54) In Example 5, the subject matter of any one or any combination of Examples 1 to 4 may optionally be configured such that the hygiene cycle management circuit is further configured to control the pumping of the hygiene agent during each cycle of the hygiene cycles.

(55) In Example 6, the subject matter of Example 5 may optionally be configured such that the hygiene cycle management circuit is configured to time the pumping of the hygiene agent and end the pumping of the hygiene agent when a specified time interval expires.

(56) In Example 7, the subject matter of any one or any combination of Examples 1 to 6 may optionally be configured to further include a power source including one or more batteries.

(57) In Example 8, the subject matter of Example 1 may optionally be configured such that the control circuit further includes a power management circuit coupled to the power source to monitor for a need to replace or recharging the one or more batteries.

(58) In Example 9, the subject matter of any one or any combination of Examples 1 to 8 may optionally be configured to further include the box being a portable food delivery box.

(59) In Example 10, the subject matter of any one or any combination of Examples 1 to 9 may optionally be configured such that the box has a flat bottom, and the spray nozzle is configured to be incorporated into a center of the bottom.

(60) In Example 11, the subject matter of Example 10 may optionally be configured such that the box is a rectangular box having the flat bottom, a flat lid, four flat sides each coupled between the flat bottom and the flat lid when the lid is closed. The box includes one or more hinges connecting between the lid and one side of the four sides and a locking mechanism configured to lock the lid in

a closed position.

(61) In Example 12, the subject matter of any one or any combination of Examples 1 to 11 may optionally be configured such that the hygiene device is configured to be detachably attached to the box.

(62) In Example 13, the subject matter of Example 12 may optionally be configured such that the hygiene device includes a container unit including the agent container, a controller unit including the diaphragm pump, the control circuit, and a battery, tubing coupled between the container and the controller box to provide for the fluid communication between the agent container and the pump, and tubing between the controller unit and the spray nozzle to provide for the fluid communication between the pump and the spray nozzle. The container unit and the controller unit are each configured to be detachably attached to the box.

(63) In Example 14, the subject matter of any one or any combination of Examples 1 to 13 may optionally be configured such that the agent container comprises a refillable bottle.

(64) In Example 15, the subject matter of any one or any combination of Examples 1 to 14 may optionally be configured such that the hygiene cycle management circuit is configured to initiate the hygiene cycles according to a specified schedule.

(65) In Example 16, the subject matter of Example 15 may optionally be configured such that the hygiene cycle management circuit is configured to initiate the hygiene cycles periodically.

(66) In Example 17, the subject matter of any one or any combination of Examples 1 to 16 may optionally be configured such that the hygiene cycle management circuit is configured to initiate each cycle of the hygiene cycles in response to a user command.

(67) In Example 18, the subject matter of Example 17 may optionally be configured such that the hygiene cycle management circuit is configured to receive the user command through the communication circuit.

(68) In Example 19, the subject matter of any one or any combination of Examples 11 to 18 may optionally be configured such that the hygiene cycle management circuit is configured to confirm that the box is empty and that the lid is closed and to start spraying a hygiene agent upon the confirmation.

(69) In Example 20, the subject matter of Example 19 may optionally be configured such that the hygiene cycle management circuit is configured to transmit a confirmation request signal for confirming that the box is empty and that the lid is closed, to receive a confirmation response signal confirming in response to the transmission of the confirmation request signal, and to cause the spray nozzle and the pump to start spraying the hygiene agent upon receiving the confirmation response signal.

(70) In Example 21, the subject matter of any one or any combination of Examples 1 to 20 may optionally be configured such that the monitoring circuit is further configured to notify the user the status of each hygiene cycle of the hygiene

(71) In Example 22, the subject matter of Example 21 may optionally be configured such that the monitoring circuit is further configured to notify the user to perform one or more functions during each hygiene cycle of the hygiene cycles according to a hygiene procedure.

(72) In Example 23, the subject matter of any one or any combination of Examples 1 to 22 may optionally be configured such that the monitoring circuit is configured to record information including one or more of information identifying the box, information identifying the user, and timing information for each hygiene cycle of the hygiene cycles.

(73) In Example 24, the subject matter of any one or any combination of Examples 1 to 23 may optionally be configured such that the monitoring circuit is configured to transmit the recorded information out of the hygiene device using wireless communication technology.

(74) In Example 25, the subject matter of any one or any combination of Examples 1 to 24 may optionally be configured to further include a user interface device configured to present information related to each cycle of the hygiene cycles to the user and receive information related



to the each cycle from the user.

(75) In Example 26, the subject matter of Example 25 may optionally be configured such that the user interface device comprises a mobile device.

(76) In Example 27, the subject matter of Example 26 may optionally be configured such that the mobile device comprises a cellphone and a hygiene application installed on the cellphone. The hygiene application allows the user to participate in control of each cycle of the hygiene cycles.

(77) In Example 28, the subject matter of any one or any combination of Examples 25 to 27 may optionally be configured such that the user interface device is configured to allow the user to initiate a cycle of the hygiene cycles.

(78) In Example 29, the subject matter of any one or any combination of Examples 25 to 27 may optionally be configured such that the user interface device is configured to allow the user to start spraying of the hygiene agent during each cycle of the hygiene cycles.

(79) In Example 30, the subject matter of any one or any combination of Examples 25 to 29 may optionally be configured such that the user interface device is configured to receive the recorded information from the hygiene circuit.

(80) In Example 31, the subject matter of Example 30 may optionally be configured to further include an analysis center configured to be communicatively coupled to the user interface device via wireless communication and to analyze the recorded information and produce feedback information based on an outcome of the analysis. The user interface device is configured to relay the recorded information to the analysis center and to receive the feedback information from the analysis center.

(81) In Example 32, a method for cleaning and sanitizing a box configured to be carried by a user is provided. The method may include verifying emptiness of the box, starting an automatic hygiene cycle using a battery-powered portable hygiene device attached to the box, recording information related to performance of the hygiene cycle, and transmitting the recorded information to an analysis center for analysis and monitoring of hygiene status of the box.

(82) In Example 33, the subject matter of starting the hygiene cycle as found in Example 32 may optionally include spraying a hygiene agent into the box to produce a mist of the hygiene agent in the box to result in substantially uniform distribution of the hygiene agent on interior surfaces of the box.

(83) In Example 34, the subject matter of spraying the hygiene agent as found in Example 33 may optionally include spraying a cleaning agent.

(84) In Example 35, the subject matter of spraying the hygiene agent as found in Example 33 may optionally include spraying a sanitization agent.

(85) In Example 36, the subject matter of spraying the hygiene agent as found in Example 33 may optionally include spraying a cleaning and sanitization agent.

(86) In Example 37, the subject matter of spraying the hygiene agent into the box as found in Example 33 may optionally include spraying the hygiene agent into the box using a single nozzle positioned at a bottom of the box.

(87) In Example 38, the subject matter of any one or any combination of Examples 32 to 37 may optionally further include using a user interface device to communicate with the user regarding the hygiene cycle.

(88) In Example 39, the subject matter of using the user interface device as found in Example 38 may optionally include using a cellphone carried by the user.

(89) In Example 40, the subject matter of transmitting the recorded information to the analysis center as found in any one or any combination of Examples 38 and 39 may optionally further include transmitting the recorded information through a network using a wireless communication technology.

(90) In Example 41, the subject matter of Example 40 may optionally further include producing instructions for the user based on the analysis and monitoring of hygiene status of the box and

presenting the instruction for the user using the user interface device.

(91) In Example 42, the subject matter of any one or any combination of Examples 38 to 41 may optionally further include verifying emptiness of the box comprises instructing the user to verify that the box is free of solid or liquid waste resulting from food delivery. The box is a food delivery box.

(92) In Example 43, the subject matter of verifying emptiness of the box as found in Example 38 may optionally include instructing the user to verify the emptiness of the box.

(93) In Example 44, the subject matter of starting the automatic hygiene cycle as found in any one or any combination of Examples 38 to 43 may optionally include starting the automatic hygiene cycle in response to a command transmitted from the application upon verification that the box is empty.

(94) In Example 45, the subject matter of starting the automatic hygiene cycle as found in Example 34 may optionally include starting the automatic hygiene cycle according to a specified schedule and upon verification that the box is empty.

(95) In Example 46, the subject matter of starting the automatic hygiene cycle as found in Example 45 may optionally include starting the automatic hygiene cycle on a periodic basis and upon verification that the box is empty.

(96) In Example 47, the subject matter of any one or any combination of Examples 33 to 46 may optionally further include starting, timing, and ending the spray of the hygiene according to a specified time interval.

(97) In Example 48, the subject matter of Example 47 may optionally further include using the user interface device to instruct the user to dry interior surfaces of the box upon completion of the hygiene cycle.

(98) In Example 49, the subject matter of any one or any combination of Examples 33 to 46 may optionally further include using the user interface device to instruct the user to dry interior surfaces of the box upon completion of the hygiene cycle.

(99) In Example 50, the subject matter of any one or any combination of Examples 33 to 49 may optionally further include using the user interface device to instruct the user to clean exterior surfaces of the box periodically.

(100) This application is intended to cover adaptations or variations of the present subject matter. It is to be understood that the above description (including the list of examples 1-50) is intended to be illustrative, and not restrictive. The scope of the present invention should be determined with reference to the appended claims, along with the full scope of legal equivalents to which such claims are entitled.

## Claims

1. A system for controlling a hygiene device configured to be affixed to a box to conduct hygiene cycles each including spray of a hygiene agent into the box and to record information related to performance of the hygiene cycles, the box configured to be used by a user for delivering food, the system comprising: a mobile device configured to: communicate with the hygiene device wirelessly; communicate with the user regarding initiation of each hygiene cycle of the hygiene cycles; instruct the user to verify emptiness of the box upon the initiation of the each hygiene cycle; receive from the user a verification that the box is empty; trigger the spray of the hygiene agent during the each hygiene cycle upon receiving the verification from the user, and receive the recorded information related to performance of the each hygiene cycle from the hygiene device.
2. The system of claim 1, wherein the mobile device is configured to allow the user to participate in control of the hygiene cycles using a hand-free operation.
3. The system of claim 2, wherein the mobile device is configured to allow the user to initiate a hygiene cycle of the hygiene cycles.

4. The system of claim 3, wherein the mobile device is configured to allow the user to override a schedule programmed into the hygiene device by initiating the hygiene cycle.
5. The system of claim 1, wherein the mobile device is configured to detect each failure in complying with a hygiene procedure established to ensure food safety.
6. The system of claim 5, further comprising an analysis center configured to communicate with the mobile device via a telecommunication network, to receive the recorded information, to analyze the recorded information, and to produce feedback information based on an outcome of the analysis, and wherein the mobile device is configured to send the recorded information to the analysis center and to receive the feedback information from the analysis center.
7. The system of claim 6, wherein the mobile device is configured to send the recorded information to the analysis center in response to a detection of failure in complying with the hygiene procedure.
8. A method for controlling a hygiene device configured to be affixed to a box to conduct hygiene cycles each including spray of a hygiene agent into the box and to record information related to performance of the hygiene cycles, the box configured to be used by a user for delivering food, the method comprising: controlling the performance of each hygiene cycle of the hygiene cycles using a mobile device configured to communicate with the hygiene device wirelessly, the controlling including: communicating with the user regarding initiation of the each hygiene cycle; requesting verification of emptiness of the box upon initiation of the each hygiene cycle; and triggering the spray of the hygiene agent during the each hygiene cycle upon receiving a command indicating that the box is empty; and receiving the recording information related to the performance of the each hygiene cycle transmitted from the hygiene device to the mobile device, the recorded information including information indicating whether a hygiene procedure is complied, the hygiene procedure established for at least one of ensuring food safety or meeting regulatory requirements.
9. The method of claim 8, further comprising detecting a failure in complying with the hygiene procedure using the mobile device.
10. The method of claim 9, further comprising transmitting the recorded information from the mobile device to an analysis center by uploading data representing the recorded information to a network.
11. The method of claim 10, wherein transmitting the recorded information to the analysis center comprises triggering the transmission of the recorded information by an end of the each hygiene cycle or on a periodic basis.
12. The method of claim 10, wherein transmitting the recorded information to the analysis center comprises triggering the transmission of the recorded information by a detection of failure in complying with the hygiene procedure.
13. The method of claim 8, further comprising allowing the user to override a schedule programmed into the hygiene device by initiating a hygiene cycle of the hygiene cycles using the mobile device.
14. The method of claim 8, further comprising presenting food safety tips for home delivery of cooked food using the mobile device.
15. The method of claim 8, further comprising displaying a training video on a hygiene procedure for home delivery of cooked food using the mobile device.
16. The method of claim 8, further comprising: transmitting the recorded information from the mobile device to an analysis center by uploading data representing the recorded information to a network; and detecting a failure in complying with the hygiene procedure using the analysis center.
17. The method of claim 8, wherein requesting verification of emptiness of the box comprises instructing the user to verify the emptiness of the box.
18. A non-transitory computer-readable storage medium including instructions, which when executed by a mobile device, cause the mobile device to perform a method for controlling a hygiene device configured to be affixed to a box to conduct hygiene cycles each including spray of a hygiene agent into the box and to record information related to performance of the hygiene

cycles, the box configured to be carried by a user for delivering food, the method comprising: communicating with the hygiene device wirelessly; controlling the performance of each hygiene cycle of the hygiene cycles, including: communicating with the user regarding initiation of the each hygiene cycle; instructing the user to verify emptiness of the box upon initiation of the each hygiene cycle; receiving from the user a command indicating that the box is empty; triggering the spray of the hygiene agent during the each hygiene cycle upon receiving the command from the user; and receiving the recording information related to the performance of the each hygiene cycle from the hygiene device.

19. The non-transitory computer-readable storage medium of claim 18, wherein the method further comprises detecting a failure in complying with a hygiene procedure.

20. The non-transitory computer-readable storage medium of claim 18, wherein the method further comprises relaying the recorded information to an analysis center by uploading data representing the recorded information to a network.

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